

Preparing and Calculating Molarity

PhET Concentration · PhET

Senior Secondary (Years 11-12)

~30 minutes

Science · Chemistry

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/concentration>

Curriculum links: solutions · stoichiometry

Learning intentions

- I can explain how the amount of solute and solvent determines a solution's concentration.
- I can calculate molarity from given amounts of solute and solution volume.
- I can use a concentration meter to verify a calculated or estimated concentration.

Getting started

Open PhET Concentration. Choose a solute and use the controls to adjust the amount of solute and the volume of solvent, watching how the colour intensity of the solution and the concentration meter respond.

Tasks

1. Add a small amount of solute to a fixed volume of solvent. Describe how the colour intensity of the solution changes as you add more solute.

2. Set the solute amount and solution volume to specific values (e.g. 0.5 mol in 0.5 L). Calculate the expected molarity (mol/L) before checking the simulation's concentration meter.

Working:

3. Check your calculated answer against the concentration meter. Did they match? If not, check your working.

4. Halve the volume of solvent while keeping the amount of solute the same. Calculate the new expected molarity, then verify it with the meter.

Working:

5. Test three different solute amounts at the same fixed volume. Record the calculated and simulation-shown molarity for each.

Amount of solute (mol)	Calculated molarity	Meter reading

6. Explain, in terms of particles per unit volume, why adding more solvent to a fixed amount of solute lowers the molarity even though the total amount of solute hasn't changed.

7. A target concentration of 1.5 mol/L in 0.75 L is given. Calculate how many moles of solute you would need, then use the simulation to prepare this solution and check your working.

Working:

Extension (optional)

If you needed to dilute a concentrated stock solution down to exactly half its original molarity, explain how much extra solvent you would need to add to a fixed starting volume, using your own working rather than just the simulation.
